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*** START OF THE PROJECT GUTENBERG EBOOK GLEANINGS IN
BEE CULTURE, VOL. III. NO. 3 ***

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PUBLISHED MONTHLY, AT MEDINA, OHIO,
BY A. I. ROOT & CO.

In the Preparation of this Journal the following are the Principal Periodicals Consulted:
American Bee Journal. Clarke, and Mrs. Tupper.

Bee-Keeper's Magazine. *King.*

Bee World. A. F. Munn & Co.

[Also Bound Volumes of the former since 1860, and Files of all other *Five Journals* that have been Published in America.]

American Agriculturist,

Prairie Farmer,

Rural New Yorker,

Country Gentleman,

Southern Farmer,
Scientific American,

[illegible]

GLEANINGS IN BEE CULTURE

Or how to Realize the Most Money with the Smallest Expenditure of
Capital and Labor in the Care of Bees, Rationally Considered.

PUBLISHED MONTHLY, AT MEDINA, OHIO,

BY A. I. ROOT & CO.

Vol. III

March, 1875.

No. 3

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ADVERTISERS' DEPARTMENT

SUNDRY MATTERS.

In making Bee veils use Tarleton in place of Crown Lining as advised on page 2, Vol. 1.

Mrs. Axtell writes us that we made a mistake on page 10, Jan. No., when we give her credit of extracting 3000 lbs.; that it was only about one-half that amount.

We have just received an extra nice lot of Alsike Clover seed, new crop, raised in our own vicinity, so that we know it is pure and safe. Single pound by mail in cloth bag 45c.; by express 35c., if over 10 lbs. 30c. only.

We learn from J. D. Kruschke, Berlin, Wis., that there *is* a good market for the oil from Rape, and that in fact we at present import it to supply the demand. We refer all interested in Rape culture to the little book to be had by enclosing 5c. to the above address.

Magazine reached us on the 15th, with its appearance much improved. On further consideration we have decided to keep the clubbing price with it and GLEANINGS at \$1.75, and \$5.00 for all the Bee Journals. If our readers will now excuse so many changes we will try to change no more.

Mr. Doolittle, of Borodino, and L. C. Root, of Mohawk, N. Y., have each sent us a club of 20 subscribers, Prof. Cook, and several others, nearly as many. A few more such friends and we might afford to give you a larger Journal and larger type, *without* any change in price.

Our thanks are due J. C. Colborne, of Chicago, for a description of the hive and frame used by Mr. Harbison. It is not a suspended frame, nor is it like Quinby's. We should prefer getting something more definite, and from

Harbison himself, if possible, before giving it to our readers. Dimensions of frame about 12×15.

Our mailing type goes by machinery that won't work unless 75c. be remitted once a year. Therefore look to the labels on your papers and see when your time will be out. If the paper stops coming blame the "machine" and not us; also, if the labels do not always present your account to you monthly, as it should be, drop us a postal.

We presume nearly every one of our readers, has already done something for the relief of the Kansas and Nebraska sufferers; to those who have not had the opportunity presented them, we would refer to Mr. James Vick's proposal in his Floral Guide. His arrangements enable him to give a receipt for all money, and to also show how and where it has all been judiciously expended.

Notwithstanding all that has been said about honey dealers, we have at least two men in whom we feel we can place implicit confidence; Lippincott of Pittsburg, and Muth, of Cincinnati. Mr. Muth has for many years been dealing in supplies, and we have yet to hear a single complaint of him in any shape or manner. His honey jars are very neat indeed, and what is more they will hold an honest pound, or 2 lbs., according to the stamp in the glass of each. As an instance of the magnitude of his business, we may say that he has given the manufacturers orders for 1000 gross for the coming season; customers may depend on getting goods as soon as ordered.

My bees, 50 stocks, on their summer stands, are not doing well; weather very cold, some days below zero all day. A good deal of ice in hives, with not sufficient warm weather to thaw it and dry them out. Have lost some already, and shall lose more if the weather does not change soon.

JOHN F. TEMPLE, Ridgeway, Mich. Feb. 5th, '75.

The San Diego Mountain honey does not candy although exposed to a freezing temperature in an uncovered vessel. Mr. Tweed says such is their experience, and astonishing as it may seem, such proves the case with us, while all our other honey under the same conditions, is white and solid.

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tfd KRUSCHKE BROS., Berlin, Wis.

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2t\$2 Dickinson, Cumberland Co., Pa.

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1tf Colerain, Franklin Co., Mass.



You cannot look over the back No's of GLEANINGS or any other Periodical with satisfaction, unless they are in some kind of a Binder. Who has not said—"Dear me what a bother—I *must* have last month's Journal and it's no where to be found." Put each No. in the Emerson Binder as soon as it comes and you can sit down happy, any time you wish to find any thing you may have previously seen even though it were months ago.

Binders for GLEANINGS (will hold them for four years) gilt lettered, free by mail for 50, 60, and 75c, according to quality. For table of prices of Binders for any Periodical, see Oct. No., Vol. 2. Send in your orders.

A. I. ROOT & CO., Medina, O.



Vol. III.	MARCH, 1, 1875.	No. III.
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REMARKS ON IN-DOOR WINTERING.

BY JAMES BOLIN.



riend Novice:—While I freely endorse the most that is contained in the article on, How to Conduct an Apiary, in the Dec. No. of GLEANINGS, I will have to dissent from a few of your conclusions, even at the risk of being accused of *heresy* again.

For instance: that the position that keeping bees warm etc., will save them, will have to be given up, and, since our wintering troubles, **** nothing that has been done has amounted to a row of pins, [Beg pardon, we meant toward curing sick ones.—Ed.] except fine weather etc. Now the above may be true in the case of your bees, but I cannot think it will apply to many. I have pretty strong evidence that keeping my bees warm and quiet, *did* save them, as most of those in this neighborhood that were not wintered in warm depositories have perished with the bee disease, during the last two or three winters. Keeping mine warm has, in my opinion, amounted to a pretty long row of pins—a longer one in fact than any one would need, unless they wished to start a notion store, and even then they might be overstocked.

I do not think that I have ever claimed that cold was the *only* but merely the *main* cause of the losses that have occurred. The want of dryness and darkness in the winter depository has no doubt had considerable to do, in some instances, with the loss of bees that were housed, while disturbance, caused by taking a light in the room, looking at the bees, admitting strong currents of air, by opening the door at night, and introducing artificial heat has no doubt killed more bees than anything else, except cold. A prominent bee-keeper remarked last spring, that he regarded artificial heat as being

indispensable in wintering bees; right in the face of the fact that he had lost about eight-ninths of his bees by its use, or at least, while using it. Rather a poor argument in its favor, I think. But I may be too practical in my views. Bees are very sensitive, and a slight jar, taking a light into the room etc., will often excite them to an injurious activity. I have frequently seen the advice given to open the door of the winter depository at night to cool and purify the air, but if the ventilators are arranged as they should be, I would much rather depend on them and keep the door shut.

You remark on page 139 of Dec. GLEANINGS, that opening the door and windows of your cellar only seemed to make the bees warmer. It no doubt had just that effect, as the bees were stirred up by feeling a current of air different from that in the room, [but what *should* we have done?—Ed.] and strong stocks, when disturbed, generate an immense amount of heat. I had a pretty fair sample of what they can do in that line two years ago when I put my bees in the house. We had a cold south-west wind at the time. The thermometer stood at about zero in the open air, and at 34° in the house when I began carrying the bees in. By having the door open it sunk to 20° in the house, by the time I had them in. I put in 88 swarms,—then shut the door for two hours,—when I went in again and they had run the thermometer up to 45°, being 11° higher than it was before the door was opened to put them in. It remained about the same all the time out of doors. By letting them alone, they soon became quiet and the temperature of the room fell to about 40°. Keeping bees *too warm* will excite them, and will have the same effect as keeping them too cold, cause them to fill themselves with honey, and if the excitement is kept up long, the result will be the same—they will be effected with the dysentery.

The thermometer in my bee house stands at 42° at this date, Dec. 14th, 1874, and a person on entering the room would almost think there was not a live bee in it, they are so still.

That the Editor, and all his readers, may succeed in carrying all their bees safely through the present, and all subsequent winters, is the wish of JAMES BOLIN.

West Lodi, Ohio.

OUR OWN APIARY.



We mentioned last month that we gave the bees in the forcing house a brisk fly on the 17th, to accomplish this more effectually we removed the covers to all the hives, and this was one reason why only *four* bees remained on the sash. They saw their comrades just beneath them and of course “hopped down” among them. To avoid giving any erroneous impression we may remark that we spend the greater part of our Sundays with our books and papers in the forcing house, and on sunny days even the rest of the family find it an agreeable sitting room. Of course we do not mean to work with our bees on the Sabbath and should be very sorry to have any of our readers get such an opinion of us, yet it must be admitted that it so came about, that the *bees* instead of being allowed to rest on the Sabbath, rested six days, and on the seventh were expected to turn out and have a fly if nothing else. If they would not otherwise, we uncovered the hives etc., as above. Very likely friend Bolin will expect this treatment to kill them whether or no. Never mind; they would probably submit with resignation if they knew it was solely in the cause of science. On the 23rd, we warmed the house in the afternoon and made search in nearly all the hives for eggs, but none could be found. We were anxious to report brood in Jan., in our Feb. No., and so kept the room at a favorable temperature all day the 24th, and next day were delighted to find that the Queens had laid profusely, even to the lamp nursery which contained less than 100 bees with the Queen at this time. By the way we can see no difficulty in wintering any number of Queens with a tea-cupful of bees each, in this way. Our last form was to be printed on the 27th, and to get larvæ before this time required careful work, so we kept the room warm until the last item was set up, but “not a larvæ” could we see, and we dolefully sent you your papers with hardly a parting note in regard to our experiment. On the 28th we looked again and were cheered with the

sight of whole patches of larvæ, so large it seemed we must have been hasty the day before. And now for pollen. Few of the colonies had any at all, some of them positively none, and the worst of it was the bees would take no notice of the spot where they had worked so industriously on it in Dec. They seemed to take their flights close under the glass. On the 31st we were rejoiced to see the sun come out full and clear, and by 10 o'clock the room was abundantly warm without any aid from the stove, but not a bit would they notice the meal. We had read in the *Fruit Recorder* that plants, especially strawberries, must be placed *close up to the glass* to thrive. Was it not so with bees? In a twinkling the Simplicity hive cover containing the heap of meal was suspended from the sash within about a foot of the glass. Our better half here interposed that it was long past church time and we bent our steps churchward with about as much alacrity as did we when in the woods with the wild Touch-me-nots. We were late, and what is more were rebuked by having the minister pause in his discourse until we could get seated. Never mind we deserved it and will try and do better next time. May our path through life never lead, where the influence of such as he is unknown.

Do you wonder that we were in a mood to rejoice more fully, with the bees, when we found them happy as in June, dancing about our heap of meal, and now and then scampering into their hive as fast as their padded legs could carry them. Yet this was a cold wintry day outside, and the sun scarcely thawed the snow on the south roofs of the houses.

We should have said before, that our second sash were put in place about the middle of Jan. It has not yet been closed up warm and secure by any means, but it does excellent service in keeping the ice from the glass and preventing the attendant drip.

Feb. 1st—To-day we had sunshine again, and brood rearing is going on beautifully. The only drawback is the drunken bees that blunder about and finally fall on the floor; these we gather up in the evening and put in the lamp nursery where they revive and at least a part of them go through the same programme next day. Not all, however, for the lamp nursery is getting built up thereby. We forgot to mention that one colony was found Queenless; search showed her dead in bottom of hive. As she looked natural and perfect, and as the bees were healthy, we cannot think it any fault of our own that we now have 67 colonies instead of 68.

[The following is from A. I. Root personally.]

Feb. 8th—I hope and trust that I have many warm friends among our readers, many who have followed my efforts, in years past, and very likely who feel that they know my weak points almost as well as those who have held daily converse with me, face to face. If I have gained any hold upon you, and if you have any confidence in my truth and candor, please listen, and do not turn away, even if I talk a little to you on these pages, on something, that does not *directly* pertain to Bee Culture.

For a little time back a great light has been breaking above me. This light might have come sooner had it not been for several things which stood in its way; prominently among them, a vain pride and ambition in regard to this very GLEANINGS. I worshipped worldly things first, and my Maker, (when I worshipped Him at all) afterwards. In fact when this great light commenced to reveal itself, I debated whether it was best to mention the matter at all, here; whether it would be—well, *profitable*. When these thoughts arose, the old darkness threatened to come back, until I could truly say, “I will do my Creator’s work first *whatever* it may be, and bees and all else afterwards.” Dear readers do you know that this is one of the first tasks shown to me, to use my influence whatever weight it may have, in all possible directions, to induce imperfect mankind to say with me *Thy will* our Heavenly Father, not ours be done.

Do you say you have no duties that you are aware of, left unperformed? just as I did a very few days ago? Go read your Bible, read the commandments and see. When you have tried to live up to these, when you have tried to love your neighbor *as yourself*, and find you *cannot* do it alone, admit your helplessness and call on your Heavenly Father for aid, but first be sure you can freely give up all or *everything* in this world for His sake, and forgive all your fellow beings, as you hope to be forgiven. With a sincere prayer that God will enable these few words to reach you just as they were intended, I still remain *more than ever* your old friend A. I. Root.

Feb. 12th—Reports come in from all sides in regard to the extreme severity of the weather, and brood rearing in the forcing house is again suspended on account of the thick coat of ice that covers the sash and prevents even the noonday sun from penetrating and warming it up. Instead of feeling like being dissatisfied with such weather ought we not rather to

take it as a lesson that our climate is uncertain, and that we should in building our wintering houses, cellars etc., make proper calculation for such extremes.

Feb. 15th—Still zero weather. The forcing house is so completely covered with ice that even the noonday sun scarcely has an effect on it. In *Jan. Am. Agriculturist*, Mr. Quinby describes the behavior of bees in cold weather, and also reiterates the statement made last season that the solid portion of the honey they eat during winter is evacuated in a dry state, and may be found on the bottom board of the hive, when the bees are in health. As soon as the paper was received we commenced some experiments to determine (as we supposed) the truth of the matter. We soon decided that Quinby was utterly wrong in both, and prepared to write a severe criticism. We are sincerely glad we did not for the spirit that was then prompting, was more a disposition to show that Mr. Q. was in error, than to get at the truth *whereever* it might lie. What we did was to raise a hive up from the bottom board, remove cover and quilt and subject them to severe cold weather. Although the colony (nucleus rather) contained not more than a quart of bees, they seemed to bear this without detriment. A sheet of white paper was placed under the cluster, and after a few hours the brown particles that had accumulated were examined. We thought then there was nothing there but bits of comb, propolis etc., but a more candid examination since has convinced us that, in some hives at least, the bees do void their excrement in a dry state, and perhaps they always do in perfect health. The second point was to see how the bees behaved when it was cold; strange to tell they did not behave at all. They were simply, perfectly still, “dead as door nails,” as Gallup used to say. We approached on tip-toe, and examined them by day light, and by lamp light, but it was all the same. We fixed our eye on a single bee, and watched it until our teeth chattered, but it seemed perfectly comfortable on the outside of the cluster. When the temperature became lower, quite a hum came apparently from the center of the cluster, but we could see no movement that should produce this; the bees that were visible, did not move their wings, and did not change places. Now then, when do these bees get food, and if they change about, why could we have not seen *just one* in the act of so doing? We confess we do not know, will some one else help? During the experiment once or twice, a bee would crawl out of the cluster and fly off in the cold and fall down and die. We then took a distant position, and saw the same phenomena, and from the number of

bees found scattered about, we think it occurs about the same whether they be disturbed or not. A bee that is sick crawls out of the cluster, and out of the hive if he can, and dies. As they die thus, most in the forcing house, we may infer that brood rearing aggravates the trouble, or what is more probable *to us* is, that sudden and wide changes of temperature, such as we always have in the spring are severe on bees as well as on vegetation. The forcing house varies from 40 to 70°, now, almost daily. If this be true, our bees had better be kept in doors until April, or even later, if we can manage to do so; and those using the cold frames, should keep them covered and dark, except at intervals, until the days get pretty warm. All colonies have a pretty fair patch of brood, in the forcing house, it is true, but the old ones are dying off so fast, we fear we are gaining little.

Feb. 16th—Found Queen in lamp nursery dead on bottom. The bees looked bright and all right, and she looked natural, except that her body was somewhat distended. Our utter helplessness, in the matter is illustrated in the following:

During the last three winters, I have suffered heavy losses, and the matter has been a great puzzle to me. My reports of the last three seasons would have been much better, had my bees wintered well. I have for the last six winters kept my bees in an exceeding dry cellar, with an average temperature for the whole time of about 37°. For three winters they did well, then came disaster. To my mind, none of the causes and methods or theories advocated cover the whole ground or seem absolute remedies for this fatality in wintering. I firmly believe that it was an epidemic (or perhaps more properly an “apidemic”) sent by Providence for purposes His own. The most curious part in my experience is that stocks so nearly alike that I could detect no *slight* difference in quality, were affected so differently—one dying or becoming very weak while the other wintered in fine condition.

J. H. NELLIS, Canajoharie, N. Y. Feb. 10th, '75.

REPORT FROM L. C. ROOT.



DITORS GLEANINGS:—You ask me to report results of the past season. I started last spring with 100 swarms, thinking this about the number one man should attend to, and considering it about as many as should be kept in one place, especially as there are over 150 swarms within one mile of us.

After placing bees on their summer stands which I did about the middle of April, my first business was to remove all combs except those occupied by the bees. Number of combs left in would vary from 3 to 7 giving space in hive to be kept warm according to quantity of bees. And here let me say that no one thing helped me more towards success, than did the Quinby frame, by the use of which I could contract or enlarge space in brood chamber at pleasure.

The yield of honey from willow, apple, raspberry, and clover was light. My principal business up to the first of July was taking combs filled with brood from strongest swarms to help weak ones, and filling their place with empty combs. Basswood commenced blossoming about the 20th of July. Then came our flush. I increased my stock to 123 swarms.

Whole amount of box honey	3000	lbs.
Whole amount of extracted	7271	lbs.
Total	10271	lbs.

Have in winter quarters 121 swarms. So far they seem to be in fine condition.

Mohawk, N. Y. Feb. 1st, 1875.

QUINBY HIVE.

The following in regard to the Quinby hive is just at hand.

You say on page 15 that “in justice to ourselves we would smile to see the person who would dare undertake to find the Queen or extract the honey from a dozen Q. hives as quickly as we could from a like number of our suspended frame hives.” I would smile to see yourself, or any one else undertake to do it one-half as quickly as some half dozen I could name. Again, on same page, “where one has plenty of leisure Mr. Quinby’s cheap hive has many advantages, and we may be mistaken about the time needed by an expert to open and close these hives.” Now would it not be candid to say that Mr. Hetherington, Mr. Elwood, Mr. Van Deusen, and Mr. L. C. Root, have as much sense as common folks. Now what can be the object of using said hive, when hundreds have to be looked after if there was nothing to be gained from it. It is necessary for Mr. H. to work fast, if it is for any one. It is his business. He has used box, straw, and modification of Langstroth—has a thousand dollar’s worth on hand, that he does not use, now. To suppose that he has thrown aside all this property without being quite sure he can work with greater facility with the hive he is using is paying his judgment a poor compliment.

M. QUINBY, St. Johnsville, N. Y.

It seems to us Mr. Q. writes a little unkindly, but perhaps we deserve it. In a matter of so much importance there should be no arguing, and no strife. Even should there be a test trial of the two hives, made by two experts, the result would be of little use to our rising bee-keepers. The question is, how will the people at large succeed best. At present we really know of no better way than for those who are undecided, to try one hive of each kind; what suits your neighbor exactly, may not suit you. Our having the Corners for sale, should make no difference in our opinion, and we try not to let it, but we cannot help wondering if Capt. Hetherington has ever tried a hive with these Corners.

In the *Am. Agriculturist* for Feb., the Quinby hive is described with illustrations. The frame differs a little from the one we described last month, in having the top bar also, dropped a little below the ends of the side pieces; also, the top and bottom bars are both alike and lighter. Dimensions there

given are, uprights, $11 \times 1\frac{1}{2} \times \frac{1}{2}$. Top and bottom, $18\frac{3}{8} \times \frac{3}{8}$, cut from inch boards. Ends are nailed firmly with finishing nails into top and bottom but projecting beyond them as has been mentioned, $\frac{1}{4}$ inch. The hoop iron hooks to hold the frames in an upright position, if they be used, can probably be bought cheaper of Mr. Q. than they can be made. The sides and top of the hive are made of $\frac{1}{2}$ inch boards, planed smoothly, just the size of the frames, with cleats nailed on each end to prevent warping. The bottom board is $11 \times 20 \times 1$, also cleated on under side to prevent warping. Mr. Q. says tie all together with a stout rubber cord, but it seems to us this cannot prove a very durable fastening.

SOME HOME MADE “NEW IDEAS,” SPECIALLY ADAPTED TO NOVICES.

BY R. L. JOINER.



riend novice:—Enclosed find 75 cents for the next year's crop of "GLEANINGS." I have just sewed 1874 together and I would not take \$5.00 for it if I couldn't get another. My bees are sound asleep in their pit at present, and will remain there until about "St. Pathrick's day in the mornin'." I took the precaution this fall of planking the top and sides of the pit, also of putting two doors in the front end, something as described in Dec. GLEANINGS. My observing neighbors tell me that I had better have let well enough alone, and buried them in the dirt as usual, but I thought a permanent place if as successful, would be better. If you remember, I used to have trouble about my bees swarming as soon as strong. Well as you told me, I found regular and thorough extracting a perfect remedy, but I soon had my hives jam full of brood. The two story plan was "no good," as my hives are practically *three* story ones now. I did not want to get new hives and I could see no way of building a story, like Pat's house "on behind." I solved the difficulty in this manner, I moved the hive half its width to the right or left and set another hive exactly like it, by its side, with the entrance the same way, and took half of the combs from the old stock and placed them in (without any care which hive had the Queen) and filled up each with empty combs, and run them through the height of the season in that manner, supplying the Queenless one with brood from the other as I extracted, and carefully destroying all Queen cells as I extracted each week. Of course I only practised this with Queens that were very prolific and had their hives *boiling* over with bees, and united again as soon as honey failed

and the brood was sufficiently contracted. I foresee and forebear all the objections that will be urged to this plan, and the main one will be, "Why didn't you let your Queenless hive hatch a Queen?" Well, I'll tell you, I wanted to see how much honey *I* could get from 25 stocks of bees even if some of the stocks did live in what we call out West a "double house." I don't pretend that it is any better than the "long idea" plan, except that when I wanted to contract my stock I had no "empty rooms" to carry into winter quarters. I am satisfied that the mammoth yields are from mammoth colonies. My yield is called enormous here but is small to what some report. I started with 17 colonies that had to be fed until June 15th. I increased by dividing, to 25 Queens, and gave 3 of them, double colonies as before described. I got 2150 lbs. of honey, 2000 lbs. of which was choice, and put 24 colonies into winter quarters well supplied with stores. I wasted the time of four of the best, for four weeks of the best part of the season trying to get some box honey. All I got for my trouble, was my pains, and the natural swarms, which I summarily returned, after throwing those honey boxes as far as I could send them, extracting every one of their *sealed combs* full of honey and destroying their queen cells. Let those who can, *raise* box honey, I had rather raise extracted for 4c per lb. than to wait all summer for box honey and then get none, for a dollar a pound. The boxes were put on "according to Hoyle," they had nice starters, they were tight, and all right every way only the bees would not move in. I am afraid they are the Novice breed and aren't in the box honey business. After I took off the boxes and took their honey, *didn't* they work though?

Well Novice, I've spun this yarn long enough now and am not half done, if you get tired reading why throw the whole away. If I ever come within fifty miles of Medina I am going to stop and see you and bore you worse than I do by letter. I'll tell you how I sold my honey.

Wyoming, Wis. Dec. 16th, 1874.

Don't go to the expense of coming here, friend J., it would hardly pay you we fear, but do keep on giving us just such sketches from your Apiary. We *do* believe you have hit on a plan that will prove many times quite practicable. For instance the Simplicity hive with, Langstroth frames, *so long as one story will hold the bees*, is to us the simplest, and easiest handled of any thing we have ever used in the shape of a bee hive, and we have studied long and earnestly in regard to some plan of uniting two of them side by side. All of these plans required too much tinkering. If we made holes for communication, through either hive, bottom board, or cover,

these holes would have to be plugged up at other times, and would look ungainly. Your plan of using them without other means of communication, than through the entrance, we confess is novel and so far as your experiment is concerned, seems quite practicable.

He who shows us how we can keep pace with modern improvements, and still keep the hives we have already in use, is truly a benefactor. To use the Simplicity hives thus, both entrances should be turned to the south, and the two hives placed close together. The covers in this case should be hinged to the front so as to turn up against the grape vine trellis, or hinges may be dispensed with entirely. In the height of the season, both hives can be pushed, well forward over the bottom boards, thus making the entrances the whole length. Should this prove "too much entrance," bank sawdust up by the outside corners. In making so many new colonies last season, we governed the size of the entrances to exclude robbers, almost entirely with sawdust, and it answered the purpose more completely to our satisfaction, than any other plan we have ever used. Also, when the nights became cool, we banked sawdust clear around the hives, to close the cracks between the hive and bottom board. We are a great friend to sawdust; it keeps down the weeds, gives you a clean place to work, is clean and orderly for a small dooryard for the bees, and enables you to make just such an entrance as you desire. Beg pardon we forgot *sawdust* wasn't our topic.

Our Standard hive with its permanent bottom board without cracks, and its capacity for 18 or 20 combs without any fussing etc. etc., would perhaps be best, but then, "we haven't got 'em," as friend J. says, and just at present we rather prefer to use the hives we have already in use.

DIMENSIONS OF FRAMES ETC.



R. NOVICE & CO:—Arn't we greenies glad that we settled down on *our own Standard* frame before you *decreed* something that one must print the dimensions with chalk over his work bench lest he forget it. We dislike the everlasting vulgar fractions and odd and even numbers about the size of frames. We adopted 12 inches inside of frame—144 square inches of emptiness—until filled with comb. Now any body can remember this size—and every one can build outside their frames just what kind of hive he pleases. It is just the nicest frame for taking in one hand, and for turning this way and that way for inspection and work, and also for leaving one hand to do something else, and we do find plenty for the free hand to do—all it *can* do sometimes. Then for shipping, why they are just the nicest and handiest frames ever made, and for housing too, hives of 10 frames pack away like brick in a wall. So you see we are not a bit sorry for ourselves—and we are so glad to have the ladies with us, we have Mrs. Tupper who is a representative character and carries the women with her in this matter. Of course we are sorry your wind-mill won't work for us in the wood work of our hives at the "Standard prices" but we can't be driven by wind into abandoning our 12×12—"On this firm rock etc." Why our hives are so jealous they won't let one frame of the Factory of the Wind, enter their yard now—we have no doubt that to set down one of your vulgar fraction and odd number frame hives would raise a mighty buzz and set all the 12 inches on a revolution against any such vulgar innovation. By the way we have got a hive "the likes on't you never did see." We took the Alley hive—paid the right to it too—and now we have some Alley, and some Quinby, and some of our own whims, and it is going to be just the nicest and handiest of a' the bee gums that can be found in Patentdom or out of it. May be we'll tell you what it is some day—if you want us to bother you.

We—that is we greenies—use the Isham honey box—the nicest, staunchest, and handiest honey box we ever saw. We know him he lives round here, he *has* a patent on his box—we saw his papers, all reg’lar—he deserves them too—has lately sold three counties to an apiarian who has seen a good many such affairs, and sold for a handsome sum. If you saw it, you would want a State at least—the *work* would suit your taste.

Now to finish all this rigmarole and to convince you that we are your friend, tho’ we expect you will blow us with a stiff breeze, we say—we frankly confess, that we have tried your tin corners for frames, after using wooden ones, and we don’t want to see in our hives another frame that has not those same tin corners. And our *green* advice to all beginners is, don’t make a single frame till you get these corners. We shall get rid of all our frames which have not these corners just as fast as we can. We have tried several hundred of them and now we send our orders for 6500 more *with the cash*, as evidence that we mean what we say.

2. EROAPPIANS.

[We can give full address to any who may desire, but the writer at present prefers a nom de plume.—Ed.]

At the risk of having the above sound something like a puff for the corners, we have decided to give it a place, especially as some disparaging remarks that have been given might tend towards giving an impression that the corners hardly deserve. We will cheerfully give place to anything on the contrary side, coming from any Apiary where they have been considerably used.

Our friends are in error about Mrs. Tupper using a frame 12×12 *inside*. Variety of taste has dictated so many different thicknesses of the stuff composing the frames, that we cannot see how our friend gets rid of fractions by using *inside* dimensions as a standard. We have seen end bars in use *one inch* in thickness and top bars with a massive comb guide still thicker. Supposing we all should adopt a uniform size inside of frame, where would be the outside, and what would be dimensions of hive to suit? We can readily make metal cornered frames to agree in outside dimensions, with any frame in use, and this allows them to be worked in the same hive mixed up with the old ones, thus giving the owner a fair chance to contrast their lightness, ease in handling, and at the same time greater capacity for brood and honey, from the larger comb surface.

We certainly don’t want any of your friend’s “rights” but if you will send us a sample of his honey box, we will cheerfully help him sell it, if we think it meritorious. Let *him* manufacture and supply all demand, and let his patent papers protect him in so doing. Nice honey boxes should be made by machinery, and ordinary bee-keepers would prefer to buy rather than to make them.

A NOVEL IDEA IN WINTERING.

WITH SOME OTHER IMPORTANT FACTS.



I have used several hives with an outside case filled in between with straw or leaves, for five or six years. Also hives with double boarded sides and ends with dead air space between the boards but I fail to see that they do any better than the single boards. Last winter a small colony in a hive of $\frac{1}{2}$ inch boards wintered well without any protection except the quilt covering on top, and the entrance closed with a piece of wool when the weather was freezing. I have two small colonies (made up in the fall of my nucleus hives) in the same hives, and they have stood thus far as well as others. The only difference that I see, is that they are more apt to fly out when the sun shines brightly, and need shading.

Another experiment.—I saw it stated that empty space beneath the bees was of great advantage. Last winter I removed all the frames from the lower story of a hive and left the colony to winter in the upper story. They did well, but it was a very fine strong colony and proves nothing. This winter I have quite a weak colony wintering in the upper story and doing very well so far.

Third experiment.—Some years ago I was preparing my bees for winter by removing frames from upper story and putting on quilts. In one very strong colony the bees remained on top the frames in large numbers. I pulled off an old wool hat and placed over them and spread the quilt over that. I found the bees filled the hat and remained in it all winter. Since then I have used all the old hats I can find. I notice in some instances that half the bees of the colony are in the hat. When those in the hat get hungry, and those below get cold, how nice and easy it is for them to exchange places, much easier than to go from the outside of the cluster to the inside over or around the frames—that is, provided that is their way of doing—for instance, when one little fellow's feet get cold he goes inside to warm them, and another little fellow comes out to take his place. This I believe is the generally received theory. I noticed it so stated lately by one of our most distinguished bee masters, (M. Q. in *Am. Agriculturist*). As I never saw any such commotion among them as would necessarily be the consequence of all this changing in cold weather, you must excuse me for being somewhat skeptical about. Has anyone *actually observed* this continual changing places in cold weather? Akin to this is another statement we frequently see made—that the colder the weather the more the bees require to eat to keep up the animal heat. This is all very nice in theory, but so far as I have observed, the bees are very quiet and still in cold weather, and scarcely consume any honey at all. These may be subjects worthy of investigation. I don't propose to discuss them here.

I have experimented some with the Adair-Gallup long idea hive, or rather “New Idea” long hive. I used two hives three feet long, one with large frames 13 square—the other with my standard narrow frames. I gave them the strongest colonies I had, and I must confess that neither of them gave as much satisfaction as the plain two story hives. They may not have been long enough (?) they did not swarm and did not fill the few empty frames I gave them.

I am aware that these experiments do not conclusively establish any particular fact or theory in bee-keeping but they may throw a little light on some points, and I find them useful in my own practice.

THADDEUS SMITH, Point Peelee Island, Ont., Can.

Friend S. it seems, has the rare good sense to see that single experiments do not settle a matter by any means. We are quite taken up with the old hat idea. Get one just large enough to hold your colony, and keep plenty of sealed honey below them, and they will be in the *best possible* wintering trim, if we know aught of bees. We once wintered a colony without any honey board. We supposed they had been given the ordinary allowance with the rest but may have been mistaken. They were out of food before March, but they had commenced to rear brood briskly—rather in advance of the rest.

SECTION HONEY BOXES.



It is at least too bothersome for *me* to make the Harbison frames, as per GLEANINGS; I have made some very nice frame boxes, top and end pieces all $1\frac{1}{2}$ inches wide, ends are 5 inches long and top and bottoms $6\frac{1}{4}$ inches long and nailed to end pieces, which are $\frac{1}{4}$ thick and top and bottom $\frac{1}{8}$ inch thick; outside sections are $1\frac{3}{4}$ wide with a groove to receive a glass 5×6 . I place the pieces in a long bottomless box or frame, wedge them up together and nail them, and then fasten strips of stout paper across them. I forgot to say in the proper place, that for an entrance I cut a notch in each side of the bottom pieces $\frac{1}{4}\times 2$ inches. I leave the bottom pieces wide because it makes a better box and is more convenient, one box can be raised up and another placed under it without much danger of killing bees, use a wax guide in each frame.

R. S. BECKTELL, New Buffalo, Mich.

[As it is a little inconvenient to cut the notches in the Harbison frame with our circular saws, we think they may be omitted and the stout paper used as above. In making these light frames, perhaps it would be well to fix on a size that would allow of putting 4 or 6 inside our large frames. Quinby advises this with his new hive, and illustrates it in *Am. Ag.* for Feb.—Ed.]

Gleanings in Bee Culture,

Published Monthly,

**A. I. ROOT & CO.,
EDITORS AND PROPRIETORS**

MEDINA, OHIO.

Terms: 75c. Per Annum.

[Including Postage.]

For Club Rates see Last Page.

MEDINA, MAR. 1, 1875.

Almighty Father! We pray that thy blessing may accompany each separate number of our little Journal on its mission among the bee-keepers of our land. May each and every one of us bear our losses during the present and ensuing months, should such there be, with a proper spirit of resignation to events we cannot control. Give us also that cool, calm, and deliberate frame of mind, that best prepares us to look earnestly about, and see what means thou hast placed within our reach to avert a repetition of our past troubles during the spring months. Give us submission, without a spirit of idleness, and teach us to work while we pray. Enable your servant to deserve the great trust, that he has never until *now* felt has been accorded him. Give him that humility that will enable him to forget self and labor honestly for the benefit of his brother bee-keepers, and to deserve the confidence they repose in him.

A. B. J., Magazine, and World, for Feb. were all on hand in fair time, and are all good.

Kruschke Bros. are now supplied with Rapp and Esparcet seeds just imported; see advertisement.

With pleasure we refer our friends wanting Queens or full colonies, to the advertiser of J. Oatman & Co.

Chinese Mustard seed seems difficult to find, but we shall have some in due time we think, even if it has to be imported.

We shall have to refer all those inquiring for bees to our advertising columns. Everything we have for sale is in our price list.

Summer Rape seed, American grown, by mail in cloth bags, per lb., 25c.; per express 15c. These will be our rates for the season.

Photo's for the Medley are still coming in, and as we wish to include all that *will* send them, we shall delay it perhaps a week after this number reaches you. Remember this is the last call, and that we wish to include every bee-keeper whose eye meets these pages.

On page 104 of Vol. 2. Mr. Quinby says the Queen of Mr. Elwood's colony that produced the great yield, *came from us*; yet in Mr. Elwood's report page 19, current Vol., it does not so appear. Who is in error? We certainly wish no credit given the dollar Queens which they have not earned.

We are happy to say Miss A. (P. G.) has recovered from the fever and is again at her accustomed post. We trust our friends will now have less cause to complain of the types and other small items, that have of late too often prevented things going with the promptness and accuracy that makes business a pleasure.

It seems some of the other sex are determined to have the "last word" in regard to bee stings; a privilege we accord them with all the good will imaginable, after we have mentioned for the benefit of beginners, that no matter how distressing are the symptoms at first, they will very soon get inured to the poison, and finally, no swelling will ensue at all.

This is the month for feeding the rye and oat meal, with most of our readers. Put in shallow boxes out of the wind, but in the sun as much as possible. Here is a chance for you to exercise your ingenuity in so arranging a glass sash, as to secure these conditions, and yet not trouble the bees by having them bump against the glass by mistake. If they don't notice the meal, as some complain they will not, start them by giving a piece of comb honey laid in the midst of the meal; get a few bees on the comb from several hives until they have got a taste, if you cannot start them otherwise. When they begin on natural pollen they will take little notice of the meal. Equal parts of oats and rye *ground fine* seems to please our bees best.

We are very sorry indeed to be obliged to say that we think "Money in the Apiary" by H. A. Burch, by no means worth the price (25c.) asked for it. The more so as our relations with Mr. B. of late have been of the most friendly nature, and he has uniformly spoken well of GLEANINGS. The book is entirely too small, containing only about one-fourth the matter of a single number of any one of our Bee Journals. King's Bee-Keeper's Text book, only costs 15c. more, and yet it is a work of 140 pages, condensed, and to the point, alphabetically indexed, etc. etc. Money in the Apiary contains less than 20 pages, less in size, and some of it unimportant matter at that. We advertised the work without seeing it which we regret, but it would be unkind, and do no good now to find fault with what is past; shall we not rather consider a remedy that will do justice all around? Our advice would be that Mr. Burch give his patrons *four* just such pamphlets, for the 25c. he has received, and make it a quarterly. If he will do this, we will give him a standing advertisement gratis.

Feb. 24th.—All three of the hives, that we are wintering out-doors, have come through so far in good condition so far as bees are concerned, although they spot their hives and the ground badly. The hive bro't from a distance worst of all; the one prepared with woolen by Miss A., next, and the Standard hive best of all. The latter stands just where, and just as it did when we left off extracting, has had no preparation nor tinkering, and is all in complete trim to extract again, so soon as its 20 combs are filled. Could we be sure of making a whole Apiary winter like this one, we should feel quite relieved, even if we *were* obliged to put two fair colonies into every hive to do it. The Queen has just commenced laying. We were agreeably surprised to find that almost every bee from these three safely regained their hives, while those in the forcing house, collect in masses on the glass, get nearly all of them into one hive, and seem in a fair way at present to become utterly demoralized, although they have reared quite a patch of brood during this month in some of the hives. Will others using the cold frames, please send us minute reports? Even though our own now looks discouraging, we are going to give it a careful test to the best of our ability, clear through.

DOUBLE WALL HIVES FOR WINTERING AND THE FINN BEE HIVE.



It will be observed that we have among our advertisements, one of the Finn Bee Hive. In receiving this, we feel it a duty to state frankly our opinion of the matter, and then if the parties think we should, we will cheerfully return the \$1.00 they have paid us for this one insertion. Of course what we say here must only be taken for our own opinion, and in the matter of selling rights we may have been unduly severe. In matters of this kind it is a very safe rule to ask ourselves if we are honestly doing as we would be done by.

If Messrs Keyes & Finn made hives for sale such as they advise and sold them for a fair price, there could certainly be no wrong, for very many bee-keepers of late stoutly affirm that bees do winter safely in hives well protected, on their summer stands, while those side by side, not so protected, perish. In order to show both sides of the question, we give the following:

They are stowed away in boxes parked with straw. Kept good in that way winter of '72 and '73, all that were packed early. Those that were packed in Jan. of same winter died, though stronger in bees and honey.

W. S. LUNT, Fostoria, O. Jan. 5th, 1875.

At present, Feb. 10th, 38° below zero. I put 28 hives in the cellar and have lost one. They are too cold, temperature varies from 29 to 34°. I would prefer 42° as nearly as possible to keep them. I put up 20 in rough boxes on summer stands, boxes six inches larger than hives, filled in with shavings. I have wintered so two years and never lost any. They seem in fine condition but it is quite expensive—cost me \$15.00 to put up the twenty hives but think it pays well. Would like to have them all put up in the same way. One of my neighbors had three stocks in box hives and has lost them all, think the loss will be quite severe in this vicinity.

Having had the pleasure of a visit to the Apiary of J. S. Hill & Sons, near Mt. Pleasant, I will, with your permission, Mr. Editor, say a few words in regard to it. Any one entering their bee yard can see at first glance that there is the best of order, system and management. It contains 85 large Langstroth hives completely storm and weather proof, adapted alike to the storms and zero cold of winter, and the broiling sun of July and August. Perfect ventilation with no possible chance of a draft of cold air on the bees, and no danger of moisture or frost in the hive to give the bees the dysentery.

Mr. Hill having years ago been convinced that cellar wintering would not do in our changeable climate, has given a great deal of thought, time and attention to the business of out-door wintering; that he has made it a success, is proven by the fact that his loss in the past four or five years, has been very light, in fact scarcely any loss at all; while others all around him lost heavily. He manages to get them through winter, strong in numbers and is not troubled with colonies dwindling down in spring and deserting their hives, as did so many the past two years where they had been wintered in cellars and other repositories. Mr. Hill is a firm believer in strong stocks—in keeping them strong the year round—takes the position that the only sure way to have strong colonies in early spring, is to see to it that they are strong in the fall. His location is not nearly as good for honey as many others, yet the amount he takes yearly would satisfy many of us who are in more favored localities; his only dependance for surplus, is white clover, having no poplar, linden, or buckwheat, consequently the honey season is rather short, ending by July 1st. The hive used by Mr. Hill is, of course, not a dollar hive, and would perhaps be thought, by some bee-keepers, to be entirely too expensive, but all things considered I believe it to be the cheapest for out-door wintering.

JONAS SCHOLL,

Lyons Station, Ind. Jan. 6th, 1875.

Were we to stop here, we might think the matter settled, but why does Quinby now advise a hive with only *half inch boards* as a protection. His large hive was most perfectly adapted for packing material on all sides of the bees, even to a thickness of 8 or 10 inches. In the large Apiaries about him would it be fair to suppose they had abandoned this plan before giving it a fair trial? We should much like to hear from Hetherington, Elwood, and others in this matter. See letter on page [29](#). The testimonials in regard to the Finn hive are none of them from practical bee-keepers such as are known through the Journals, and none of them owners of large Apiaries. If we wanted to make a hive such as they describe in their circular we assuredly should not think of buying a right for the privilege, but if they would make us a hive at a reasonable price suitable for receiving our frames, so that we could easily set a colony into one, we would prefer to buy of them rather than make one.

If Adam Grimm, and Capt. Hetherington will give their opinion in regard to double wall hives compared with single ones for out-door wintering, we

will cheerfully pay any reasonable sum for service. During our protracted cold weather, many complaints have come in, in regard to ice forming in hives left out-doors, and many losses are reported already. In cases like these, we do think the straw mats a great advantage, and is it not possible that they are as efficient as the expensive double walls?

HONEY COLUMN.



riend Root:—Please name a few parties of whom I can buy Basswood honey and oblige.

CHAS. F. MUTH, Cincinnati, O. Feb. 3rd, 1875.

I have now on hand about 60 gallons of that choice thick Clover honey, same as you bought of me, there is very little sale for it here. Price 18c., delivered here.

WM. PAYNE, Spencer, Medina Co., O. Feb. 1.

Have sold our Catnip honey to W. G. Smith. St. Louis, 15 cts. delivered, he paid promptly. The barrel came to \$64.55 net; that isn't bad is it? Shall set out half an acre of plants next spring, and sow some seed too.

J. L. WOLFENDEN, Adams, Wis.

I sent my honey to Barber & Stout, Cincinnati, O., got 15 cts. cash on delivery and was paid to a cent. I also sent J. Lippincott, of Pittsburg, Pa., one barrel. Now Novice, please accept my most profound thanks for your assistance in disposing of my honey, by your recommendations to the two above named honey men. There is nothing like keeping bees to restore a burned farm, and dollars are by far the best chromos.

J. DUFFELER, Wequiock, Wis. Dec. 16th, '74.

Humbugs and Swindles Pertaining to Bee Culture.

[We respectfully solicit the aid of our friends in conducting this department, and would consider it a favor to have them send us all circulars that have a deceptive appearance. The greatest care will be at all times maintained to prevent injustice being done any one.]



e cut the following from E. C. Hazard & Co's circular, 192 Chambers St., N. Y.

HONEY.

These packages are made of flint glass, tight, very attractive, convenient size, and salable, possessing a marked advantage over all other Honey in the market, because of its non-congealing tendencies. The Extracted Honey in bottles and tin, culled from Alsike Clover, and Orange blossoms and thrown from the comb by centrifugal force, is entirely free from all foreign substances and will be found to possess the delicate BOUQUET and medicinal qualities so seldom found in the ordinary commercial article.

E. C. H. & CO'S PURE EXTRACTED HONEY:

In Quarts, Fancy Decanters, per doz. \$4.50

We might think this a mistake but the *Grocery and Provision Review* of Jan. 18th, quotes precisely same prices. As a quart of good honey weighs 3 lbs., these good people are selling it at 12½c. per lbs. and charging nothing for those *Fancy Decanters*, packages, cost of putting up etc., etc., and yet producers get from 15 to 16.

Reports Encouraging.



can report 850 lbs. comb and 150 lbs. extracted honey from six swarms with Quinby hive.

O. J. HETHERINGTON.

East Saginaw, Mich. Nov. 25th, 1874.

From 34 hives we took 3000 lbs., 600 lbs. box and balance extracted.

T. E. HAWKINS & BRO.

New Frankfort, Mo. Dec. 18th, 1874,

Twenty-three colonies of Italian bees made 160 lbs. of extracted honey to the colony last year up to July 15th.

JOHN SCHEERER, Ridgley, Mo.

I commenced last spring with one swarm on three combs, and they were not crowded at that. I extracted 55 lbs. honey, increased to four that have 22 lbs. average of their own stores.

O. W. PARKER,

New London, Minn. Dec. 25th, 1874.

Twenty-two stocks in spring—taken 1120½ lbs. extracted honey, 288½ lbs. box honey, have now 46 strong stocks and 14 Queen rearing nuclei and weak stocks. Increased by artificial and natural swarming and lost several natural swarms.

M. PARSE. Pine Bluff, Ark. Nov. 22nd, 1874.

I had 26 stocks of bees last spring in Langstroth hives, one-half of them were very weak. I got 1100 lbs. of box honey from 20 stocks, it was sold for from 20½ to 28½ cts. per lb. I have increased my bees this year to 45 stocks which I have put into winter quarters with from 30 to 35 lbs. each.

Wm. J. DEDRICK, Borodino, N. Y. Dec. 14th, '74.

I had 60 swarms in the spring, some of them very light in bees. I increased to 100, from natural swarming, and they made a little over 5000 lbs. of box honey, including that not capped. I hived no second swarms. I used virgin Queens mostly, and cells. Often some would be about hatching; I would then put them in after a swarm issued. Bees seem to be wintering well.

D. C. McCALHOUN, Hornellsville, N. Y. Feb. 1st, '75.

32 swarms in spring, made 3000 lbs. surplus, increased to 50, (of course it was all extracted) and have sold it all for 15 cts. per lb. delivered on track. The honey was nearly all from basswood, clover did not do much. The fall was very dry and the bees got very little after basswood failed. Don't know how they will winter, but they are all right yet.

JAMES SCOTT, Epworth, Dubuque Co., Iowa.

Last winter I put up 68 swarms, all came through alive but lost 4 which were Queenless; sold two more which left me 62. Increased them the past season to 99 swarms and got 2600 lbs. box honey and 600 lbs. extracted, for which I realized about \$600.00. I sold 31 swarms for \$7.50 each, so it leaves me with 68 swarms again this winter, all of which seem to be wintering well.

W. H. TENANT, Eureka, Wis. Jan. 18th, '75.

Began the season with about 20 stocks in poor condition. Five stocks Queenless in spring. Increased them to 37 in fall, in apparent good condition to winter. From 6 stocks in non-swarmer, took 560 lbs. box honey; from the best, 110 lbs., from the poorest, 65 lbs. Built up and increased the remaining stocks, and took 760 lbs. liquid white honey from them. Fed in the fall, 15 lbs. "A" sugar. Reared during the summer, 36 surplus Italian Queens.

J. H. NELLIS, Canajoharie, N. Y.

We have only extracted from 5 stocks this summer as we thought best to "go slow and sure." Well, we took from those 5 stocks over 300 lbs. of honey, and increased them to 13 good stocks with plenty of honey for winter supplies, while our other 4 left for box honey and natural swarms, have swarmed altogether too much, and gave very little surplus. And worst of all, some of the young swarms went to the woods, in spite of all our endeavors to prevent them.

ILA MICHENER, Low Banks, Ont., Can. Oct. 19, '74.

The summer was very dry, so there was very little honey stored. But the fall was unusually fine. *Four* stocks devoted to box honey, gave 128 lbs., an average of 32 lbs. per stock. Nine stocks yielded to the extractor 741 lbs., an average of 82½ lbs. per stock. The largest yield from any *one* stock was 153 lbs. I increased my 13 to 23 and they go into winter quarters in good condition with 50 lbs. of stores per hive. I should have extracted a little closer, but was away from home at the time it should have been done.

DR. W. H. P. JONES, Nashville, Tenn.

Last year I tried small frames something like Harbison's, only I had them so that the bees could work all ways through them—could tier them up etc. It was on a Quinby hive or rather frame, put small frames on sides and top, got between 90 and 100 lbs. of comb honey and one swarm of bees besides. I shall try them several ways the coming season. The best I have done with

boxes is about the same as above. My frame is 14×10 inches, inside measure—have three New Idea hives—bees swarmed out of them while I was extracting in spite of all I could do. How's that?—never saw such a season for swarming—returned most of mine. I have only 26 swarms and do not want to increase if I can help it for I cannot attend to them. I winter in a large cellar and lose no bees, keep up my experiments winter as well as summer— that's half the fun. Most of my frames are so placed that the boxes come up plump against the *ends* of frames and are just as close to the brood as those on top and in fact the guide comb is a good deal nearer, and no bee can stick his head out at the ends, unless he sticks it in the boxes; but I forget you are no box man!

R. H. MELLER, Amboy, Lee Co., Ills.

HOW TO FLY BEES IN A ROOM.

In the spring of '73 bought two colonies of bees, having poor health thought the attention to them in the open air would be beneficial; finding the business so pleasant and profitable concluded to make it a permanent business, and have given my whole attention to it. Increased my two colonies to ten, bought 12 more, making in all 22. All came out good in the spring of '74, and I got 2100 lbs. box honey and 400 lbs. extracted, and increased to 47. I have now 65 colonies in my cellar all in good condition except one, which showed signs of dysentery. I gave them a fly, and this is how I did it. Took pine strips one inch square, made a frame 4 feet square and 2½ deep; covered sides with news-papers tacked on, spread papers on the carpet of sitting room near south window. Set my frame on it covering the top with mosquito bar, set hive outside with entrance opening into it through a hole cut in paper. Waited until bees were all quiet then warmed up room to 65°, standing thermometer against hive; all flew well for five hours, cleaned out their hive, and as darkness came on, all returned to hive again, making it a perfect success. The papers were badly soiled. Shall serve the rest the same if necessary. By flying them this way with but a trifling expense you need not lose a bee. Winter my bees in cellar with dirt walls, temperature 40 to 45°, cellar very dry. I use Langstroth hive, got most of my honey in 20 lb. boxes, two boxes cover a hive—not quite as salable but I get good deal more honey in large boxes. Box honey averaged me 24c. per lb., extracted 20c. per lb. Extracted only to give room for the Queens to lay.

D. BASSFORD, Watertown, Wis. Jan. 25th, 1875.

P. S.—Would not advise any man to fly bees in the sitting room without the full consent of his better half.

BASSWOOD OR LINDEN: STARTING A PLANTATION.



I have no Linden seed on hand at present. The seed should be planted in the fall or kept in damp sand and planted *early* in spring. As a general thing the seeds are about two-thirds bad. I think cuttings are best to raise plants from. They nearly all grow, if rightly handled. I sell the cuttings in spring at 30 cts. per 100, free by mail.

THOMAS J. WARD, St. Mary's, Vigo Co., Ind.

The above answers many inquiries. In the fall of 1871 we gathered and planted about a half peck of basswood seed, according to the directions given in Fuller's Forest Tree Culturist. So few of these came up that we never used them at all, preferring to get our 4000 for the 10½ acres from the forests. We have reason to think that basswoods grow better when partially shaded, than in the open ground. On a part of our lot, there are about 50 large white oaks; we at first hesitated about planting the young trees among these, but now find those among them, have made the best growth of all. Perhaps many of our readers have noticed the rank vigorous growth that these trees often make when young, where they stand in a thicket of bushes and briars sometimes in fence corners. Shading the ground around the roots, from the hot sun, when young, we think perhaps an important item. Although we have never tried cuttings, we think it probable they would answer excellently. A tree on one of our streets that was planted out with some maples about seven years ago, blossomed last year for the first time, but it was taken up when so large that it did not begin to grow, until about three years after transplanting. For reasons mentioned, we should advise close planting at first, say 10 or 12 feet apart, on a plan similar to the one

given in Vol. 1, pages 2 and 25, for locating hives in the Apiary. When the trees get crowded, thin out. The timber will soon pay all expenses.

Heads of Grain, FROM DIFFERENT FIELDS.



Listen to the patent right vender and do nothing but shake your head, says the *Rural New Yorker*, but I don't approve of that way. My Mother often told me when I was little that I would learn to *butt* if I shook my head so much. I will tell you how I served an "Agent," of some kind. I was in my Apiary working with my bees, and he laid his satchel on a hive that was near, and while he bothered me, he still kept knocking the hive with his satchel. The bees began to get cross, but he didn't think you know. Pretty soon the bees came out and "went for him" lively, he began to dodge and slap, but he soon grabbed his satchel and began to "beat a retreat," slapping and cracking his head with the satchel and spoiling his fine hat all to pieces, he "hollered" back he had enough of the bee business and left. It was laughable to see him "light out."

V. McBride, [page 11, Jan. No.] says that during the winter of the malady, all the bees in Langstroth hives died, and two-thirds in others etc. I saw bees in all kinds of hives that died, some in box hives from one foot to five feet high and some in some old washing machines turned up side down, others in the gable end of a house, some in the old straw hives and all kinds of patent hives, and by the way they died in all kinds, about the same. Some had the patent hives and lost their bees and then blamed the patent hive, but I found it was their fault oftentimes, as it makes it a little handier for them to divide their bees and to take their honey. They think the patent hive ought to make honey without bees almost. I guess the "*king*" don't manage right some how as some of those old farmers call them; don't you think Mr. Root, that it is the King's fault that the bees die?

D. H. OGDEN, Wooster, O.

We guess it must be the *King* friend O., for we feel sure the *Queen* is not to blame. Our very best colony in 1873, dwindled down to the weakest in the spring of '74. They went down to a mere handful, swarmed out twice,

and it was only by giving them hatching brood several times that we could barely get the Queen through until July, and then she proved herself fully equal to what she had been the season before; in fact she kept putting two or more eggs in a cell all through the spring months. It is only the *workers* that die off as soon as brood rearing commences. The very same process is now going on in our forcing house (Feb. 5th.) yet the brood will get ahead we think. Keeping the sick bees warm in the lamp nursery revives them some but they soon die nevertheless. The idea advanced that it is a kind of *fly* that kills the bees, can certainly have nothing to do with our losses.

NOW FOR THE HOT-BED.

Jan. 22nd, we let the sun in to the pit and at 2 P. M. there was a perfect jollification, just like a lot of young bees playing in summer. And they *did* “spot” things inside the pit at a great rate, showing they needed a fly. As soon as it began to cool down they all went in to the hives except some dead ones on the straw, no more than would have been found if they had flown the natural way, and about two-thirds of a tea-cupful which were stuck on the sash in groups from one to a dozen. These latter we brushed out on a dust pan and put in top of one of the hives, as they were still alive, and said hive was mighty “sassy” too; seemed strong in bees, plenty of honey, and combs dry. The pit was not dug till after the ground was frozen 8 inches deep, consequently there was considerable frost on the glass which we swept off as soon as the sun loosened it a little. The dampness inside the pit was soon dried and everything went lovely.

We were going to hoist our hat for the hot-bed, but guess we’ll wait till spring before we shout too loud.

Don’t forget the “Medley” for we want to see Katie Grimm, Mr. Grimm, and all the rest.

W. M. KELLOGG, Oneida, Knox Co., Ills.

A Mr. Abbot, of Wakeman, O., uses leather instead of Quilts. He says the leather is cheaper and does not rattle or stick. He fits them tight on the top of frames and they did not mould last winter and the bees wintered better than under quilts or tight frames. My bees in my hot-house when it has been warm enough for them to fly (no fire as yet) readily find their way back to their own hives, while those on their summer stands (though protected on the top and north sides) died by hundreds if not thousands on the snow as they lit and fell on it last week.

T. L. WAITE, Berea, O. Feb. 4th. ’75.

We had an intimation some time ago that Mr. Abbot had a remedy for propolis, above the frames. Leather being porous, might do very well, but will it not kill bees when pressed down on them? Again, can we get a piece that will cover a hive for the price of a quilt, and what kind of leather is best? Mr. Abbot claimed that he had some valuable information *to sell*, on

the subject. We presume most of you have discovered that those who claim to be possessed of valuable information, *not to be found in books*, are generally somewhat of a fraud; never mind friend Abbot, if you have by experimenting got hold of something valuable, bring it along. We will pay you for your time and trouble.

I had to-day to open the entrance full size to almost all hives. Bees keep up a lively hum and are carrying in loads of pollen equal to spring work. Considering your mortality from disease and cold in the North and your continual *trouble* and *anxiety*, I think we have far the advantage of you.

I lost one very weak swarm this winter by robbers. This is by far the worst enemy I have to contend with.

J. B. RAMSEY, Abbeville, La. Jan. 21st, '75.

And with your advantages, friend R., why do you not build up larger Apiaries, and raise honey by the car load as do our friends in California? We can imagine the smile of relief that would spread over the faces of *some* of our readers, had they no worse trouble than *robbers* to contend with.

When I ordered the two first volumes I thought I could do without GLEANINGS 3rd Vol.—But it won't DO. Don't forget the Jan. number. 'TWONT DO! No, if "blue eyes" falls down over the boards you *must stop and help her up*. In our haste, don't let us make a God of our Bees.

Have 35 stands in box to be transferred, have adopted the Standard. Shall want an extractor etc.

C. M. JOSLIN, M. D. St. Charles, Mich.

I examined several of my colonies the 20th of Jan. and found brood in all stages, from the egg to the hatching bee. One stock with brood on four combs. Lost none to date, though one hive has soiled its combs considerably.

H. PEDEN, Mitchellville, Tenn.

From several similar items, we see it is nothing very unusual to find brood even in Jan., in the Southern and Middle states, whether this is desirable or not, we are unprepared to say.

Many thanks friend "Novice" for the binder you sent me, I never expected such a nice premium. We have every number of "GLEANINGS" in its proper place in the binder, and let me tell every bee-keeper, to get Emerson's Binder for "GLEANINGS," then they can preserve them for their children's children if they like. But "Novice" I never said "confound it" if I could not just find the desired number of GLEANINGS. No indeed! we always kept them in a drawer by themselves so that we always

knew just where to find them. (Teach us order and precision and then ask such a question as per advertisement.)

“Novice” tells us to “have every comb built between one and the side of the hive” in order to have them straight, but it won’t always work unless the old combs have brood or sealed honey at the top. I have had the old combs widened out so as to fill the whole space, and nothing put in the empty frame at all. I believe this will always happen if empty frames are inserted immediately after extracting. The remedy is, use the Standard hive, move back the division board just to give room for one frame at a time, move back the combs till you come to the middle of the brood nest, insert the empty frame there, and, Ho! you will have the whole frame nearly filled, with nice worker comb, *and eggs too* before morning perhaps. This should only be done when the bees are gathering honey nicely.

Ila Michener, Low Banks, Ont., Can.

If it is only yourself that reads GLEANINGS, they might easily be kept nice “in a drawer,” but we confess that we felt rather flattered a short time since in visiting several bee friends, to find the last number at one place conveniently on a stand with a pair of spectacles laid on the pages to keep the place, and at another the number, (although it had only been out a week) looked as if some urchin had used it for a spelling book during a whole term at a country school. It certainly had been *used*, and we went our way rejoicing. Friend Ila is right about strait combs, and we thank him for the correction. But beware of spreading the brood combs however, before the weather is quite warm.

Mr. D. Dubois, Newburg, Orange Co., N. Y., sends us an ingenious plan for a movable portico to be applied to any kind of a hive. It is made of two boards 4×11 inches, fastened together at the top like a letter A. Now imagine the horizontal cross bar on this letter, a button, with a screw through the centre to hold it up firmly to the hive, and you have it all. Each end of this button is made with a projection that enters a slot cut in each of the boards, but not quite through them. By boring a hole just above the button, we have an upper entrance, with the button for an alighting board; in the winter by turning this button perpendicularly, it closes the upper entrance. To keep the portico from slipping down, a nail is driven just underneath the peak. This allows us to make the upper and lower story just alike and yet have a portico *if any body wants one*. After testing hives with, and without, for two seasons past, we really cannot think it makes any difference either way with the honey crop. Our friend Dean says a portico encourages the bees in hanging out of doors, and he wants *his* bees in the

hive. Now if the spiders should persist in making a new web in these porticos every morning, as they sometimes do in ours, we can easily lift them off and put them away. Some will probably always prefer porticos, while others will not, as in other things.

DEAR NOVICE:—Please say to the inquisitive ones, through “GLEANINGS,” that the \$34.45 mentioned on front cover of No. 6, Vol. 2, was the cost of the iron work mentioned in the same sentence, at the factory.

That the saws, bolts, belting, and lumber ran the bill for materials up to about \$48.00.

That I put the machine together myself, without adding to the figures.

That while the machine does do more accurate work, than I have ever succeeded in having done, by more portentous machinery; nevertheless, the idea of using human muscles, as the motive power, for the amount of work that looms up, prospectively, in the immediate future, does not correspond, very well, with the boasted ingenuity of this age and nation.

Therefore I am preparing to “attach” a lever horse power to the machine, by means of a belt thrown over a pulley, placed on the driving shaft.

This winter, in these parts, would do no discredit, to an arctic region. This morning the mercury fell to 40° below zero, and was still going down, when I took it in out of the cold; because it had reached (and in fact was a little below) the end of the graduated scale, at 40°. The mercury in the cellar, of late, keeps vibrating about, and near to the freezing point. Bees quiet.

D. P. LANE, Koshkonong, Rock Co., Wis. Feb. 9, '75.

How long will it take to feed 25 lbs. of sugar syrup with the Universal feeder to an average colony of bees in warm weather?

We have never tested the matter, because we deemed feeding a pint or a quart a day, more desirable in preparing for winter, than more rapid feeding. If the bags were made to hold about a quart each, and were filled morning noon and night, or oftener, we could probably get them to take 25 lbs. in two or three days, in warm weather.

How will molasses hhd. answer for holding honey temporarily? Say, take one head out—thoroughly clean—heat in the sun—then coat the inside with hot wax, using a brush for the purpose. When the honey candies there will be no difficulty in getting it out.

We would advise trying the plan you mention. If you pour in a quantity of melted wax, and run it all over the inside, it will assuredly make it hold honey.

How long before you can give us a simple and infallible remedy for honey crystalizing? Such would be worth a good deal to me as my whole crop is sold in bottles and will not retail well in a

solid state. For the present, will heating to the boiling point in a vessel set in water answer? Must the honey be bottled and sealed while hot or will it do as well when cold?

We find no trouble when we follow directions given heretofore, viz., heat your honey almost to a boil, fill the jars full, and seal *at once*, while hot, just as if it were fruit. If not made hot enough, it will candy again partially. We think the writers who say that *pure* honey will *always* candy, a little hasty.

The frames in my hives rest on a metal rabbet and are not secured at the bottom. They slide about very easily. When I send off a load of hives to an apiary, I secure the frames by placing half inch strips reaching to the bottom board between the combs at each end. It takes considerable time to fix them. Can you devise anything that will be less bother and answer equally well?

To be sure if we have frames perfectly *movable*, they must of a necessity be made stationery when we wish to transport colonies and the question arises as to whether it is advisable to be bothered with any arrangement for keeping frames fixed, every time we open a hive, just because we have once in a great while a case that makes such an arrangement desirable. In moving our bees to the swamp, see Vol. 1, page 75, we put strips between all of the end bars to the frames, except one hive which was overlooked, but as this hive had not been opened at all, the bridge of wax from one comb to the next, kept the frames all in their places. This is generally sufficient for moving short distances, where the combs have been in use several years, and where they have not been taken out for some little time before moving. In shipping bees considerable distances, we know of no plan better than the sticks. Something could be added to the hive for this when making, but would it be advisable, when the hive in many apiaries may not require to be moved in years? In moving the hives in doors and out, even with Metal Corners as well as rabbets, no preparation is needed, if the hives are carried so that the frames do not oscillate by the motion of stepping.

Are Basswood trees ornamental, and of rapid growth? Would they grow here? Our climate will admit of the culture of the hardier varieties of grapes.

G. C. MILLER, Mt. Hanley, Annapolis Co., N. S.

The Basswood is a most beautiful tree for ornament, and when it blooms the perfume extends for a great distance around. It is a very rapid grower when once started and we believe is perfectly hardy so far as frost is concerned.

Bees worked on rye flour all through the month of Dec., and most of the month of Jan., but are housed up now on account of cold. I have 68 colonies mostly in good condition, and think I will have same number when spring opens. "Long may you wave."

J. F. MONTGOMERY, Lincoln, Tenn.

DEAR NOVICE:—I am in favor of deep frames for these reasons: Experience has convinced me that bees winter better and breed earlier in deep than in shallow frames. In hollow trees: the natural home of the bee (and the one they like most) the depth is always greater than the width. In shallow frames they have to spread out so thin, or cluster on the cold honey, that it makes it much harder to keep up the necessary warmth. They seem to think they must have bees where they have honey; and I have noticed these outside guards often get chilled and die at their post. The natural place for bees to store honey, is in the top of their combs; and when they want to know what they have in store they look for it there; hence we often see them start from the cluster run over the honey as if estimating it then pass into the cluster, doubtless to report. When the honey is directly over them I think they are more fully impressed with its possessions. A hive 12×12 and 16 inches in depth I believe is the best size and shape for wintering. Such a hive will allow plenty of honey in the top and enough empty comb at the bottom for the bees to cluster on, and keep in a round form which is certainly the best. But it will be said shallow frames are handier—should we not try to please our bees as well as ourselves? Again they swing together too badly—cannot some fixture be made to keep them apart? What says Novice?

CHAS. WILKINS, Ott, Oregon. Dec. 4th, '75.

But bees die in hollow trees nearly, if not quite as badly as in hives; as new swarms generally select these vacated hollow trees, we forget that not one colony, but a half dozen may have occupied the tree in a dozen years. We know bees run over the combs, frequently, but we can hardly accept the idea that they do it exactly for the reason mentioned. We do not use shallow frames for the reason that they are handier. See pages 16, Vol. 1, and 29, Vol. 2. If hives and frames are properly made, they can be made to hang true even when 16 inches deep.

I wish to have all understand that my one dollar Queens are the poorest. My prices are for a Queen \$1.00, for a choice Queen \$2.00, for a tested Queen \$5.00.

J. W. HOSMER, Janesville, Minn.

The above was sent us by a subscriber, who had written Mr. H. in regard to \$1.00 Queens. We are alone responsible for Mr. Hosmer's name appearing in our list. He wrote us that he had been selling Queens for \$1.00 for some years; shortly after he sent us (we presume by mistake) 75c. the

second time. As he made no reply in regard to a query as to how we should use this, we took the liberty of keeping his name in during the year.

We need hardly repeat that we never intended dollar Queens, to include such as had been *tested* and found *poor*; and we do not wish to include the names of any who propose to do this, in our list.

I make a hive, 12 frames about 11×13 inches, inside measure, frames, division boards, entrance blocks, bottom and cover, with one cotton cloth feeder as you recommend, painted three coats, all complete for \$1.50. If you wish, you can say through GLEANINGS that I will give one of my hives complete for a sample to any one that will send you a club of 10 new subscribers for GLEANINGS, from Canada, (no patent on it) for 1875 by your giving them the certificate showing that they have done so.

D. A. JONES, Tecumseth, Can. Feb. 8.

Many thanks friend J. We would be very glad to offer *you* something in this case, but at 50c each there is scarcely any margin at all after paying expenses.

FRIEND NOVICE:—My hives are 15½ inches wide for ten frames; would they be any better if narrower? Will bees work as well in small frames for surplus honey as in boxes? Do you think the bee business will ever be over-done in the United States?

H. LIBBY, Lewiston, Me. Dec. 14th, 1874.

If you mean that you put ten frames in a space of 15½ inches we should advise using eleven frames instead of ten. If you mean 15½ is the length of top bar, we should consider it as good as any unless you use your hives two story, in which case a little longer frame might be better. Small frames put inside of larger ones, have often been tried but there are many difficulties; they must be made very accurately, to stay in place, the bees do not seem to like so much wood in their way, every thing is covered with propolis, and their owner generally concludes that the arrangement is too much bother to use on a large number of hives. We have just as much fear that too much butter and cheese will be produced, or too many eggs, as that the market will ever be overstocked with honey. What has been the result with small fruits? Remember too that they are perishable goods, while honey will keep safely for years.

In answer to *J. H. Irwin* in Feb. No., page 22: He can find as much comb honey in Langstroth frames as he may need, by writing Paul Dunker, Freeman, Cass Co., Mo. He has 500 to 700 frames.

W. G. SMITH, St. Louis, Mo. Feb. 4th, 1875.

I thought it was quite settled now that to raise brood, bees are obliged to have farina in some shape, but I was amazed to find what a quantity they consume, by starting my three colonies with only brood and 25 or 30 bees to a hive. I thought I was liberal in supplying them and they got on swimmingly at first, then the dead brood commenced appearing and I gave them more pollen and all went well again. I found out too that *very young bees* in an emergency like that, gather pollen as well as older ones.

ANNIE SAUNDERS, Woodville, Miss.

In '73 my bees paid in honey sold \$22.50, in '74 about \$12.00 per hive. I have had your experience for the last two years in the loss of bees.

H. W. MINER, Saranac, Ionia Co., Mich.

Excellent season for bees, had between 1500 and 2000 lbs. surplus in boxes. Sold, for from 18 to 25c per lb. Box hive man—can see no better way yet.

J. F. TEMPLE, Ridgeway, Lenawee Co., Mich.

GLEANINGS is indispensable, but my wife says I get some new hobby from every number.

WM. H. ROOT, Port Byron, N. Y.

Tell your wife that healthy, wholesome hobbies are always productive of good, and that we shall always strive to have GLEANINGS teach none other.

Can I make a bee hive and use a movable frame such as Mr. Quinby describes, without paying for the individual right? And if I have to pay for the right, who is the proper person to be paid? and how much will it cost to make and use for myself, say one or more?

T. H. APPLE, Meadville, Pa.

We are happy to be able to say that you may make hives, in any way you desire, so far as we know, without the least necessity of paying any body a right for any thing.

We got up a resolution at our convention, to apply to our Legislature for a law to label all packages of honey with the producer's name, and let the seller be responsible for adulteration, if not mentioned,—subject to a penalty when detected, etc., etc. Can you give us advice how to act?

M. QUINBY.

We do not know that we are able to give any advice in the matter. Would not the law like many others be dropped and forgotten because no one would enforce it? It seems to us that the great work is to educate consumers to know *honey* and to demand it. This is all that we have to rely on in a great variety of goods, and that establishment that once gets a name of dealing only in *genuine* commodities, has its fortune made. People are learning rapidly. If any one likes the *cheap* honey let them use it.

I have given my bees some rye flour, they take it first rate, for the last two days they have been carrying it in pretty freely. It is bolted flour such as the bakers use. Is it as good as the unbolted or not? I have not lost a colony yet and they all have sealed honey enough to spring them I think, but I have commenced to feed for brood rearing; feed syrup made of white sugar.

Now friend Novice, I want some advice. I want to adopt some other hive than the Buckeye, but I want a two story hive that I may procure, not box honey, but frame honey, and I want the frames in both stories alike so I may use them interchangeably.

T. B. PARKER, Goldsboro, N. C. Feb. 1st, '75.

The bolted flue flour, is just as good as the unbolted, except that the bees sink down in it. If mixed with bran or sawdust it does very well. When they can choose, bees always prefer rye and oats, to wheat flour.

Use the Langstroth frame by all means, if you are going to have a two story hive. If you have not seen the L. frame you had better have us send you a sample by mail. We should also use the Simplicity hives to hold them, for then you can have both stories also, just alike, and perfectly interchangeable. We can furnish a one story hive, frames, quilt, and all complete for \$1.75, and *two of these* makes a complete two story hive. We decidedly prefer the L. frames for two story, on account of their shallowness, any of the other frames loom up so tall when placed one above the other. We think we have ample evidence that the Langstroth hive winters equally as well as any deeper frame in any climate.

Our only reason for giving a preference to the Standard frame over the Langstroth, under any circumstances, is that more than 10 L. frames, placed side by side, make a hive inconvenient to make and inconvenient to handle.

Therefore if you are going to get your surplus *above* the brood combs, use the L. frame every time. The Quinby would come next to it, as it is much the same thing on a larger scale, but the American, Gallup, or Standard, are not suitable to be used two stories as a general thing. Single instances 'tis true, may sometimes seem to point otherwise, but we have *gleaned* the above from a large number of reports extending over several years. Any of the frames will we think, work very well on the plan given on another page by friend Joiner, but the last three mentioned would stand a little the most compactly.

Do you believe a pure Queen fertilized by a black drone will produce pure drones? and do you believe pure Queens fertilized by such drones, will produce pure stock, and do you allow such drones in your Apiary? What would you charge for—say 3 cards 4×5 inches each of drone comb filled with fresh laid drone eggs from your imported Queen, with bees enough to keep up the heat, and send by express in May next?

GEORGE K. HUFFMAN, Effingham, Ills.

So much time has been wasted in discussing this question that it is questionable whether it be best to hazard an opinion, until we can be *sure* of solving the question positively. If a black Queen could be fertilized, *beyond mistake*, by an Italian drone, we might readily decide the matter, for we should then have one or two banded workers, and black drones invariably. Will those who are equal to this task, if any such there be, please experiment and report. Will. R. King, of Franklin, Ky., did claim to have made such an experiment, but evidence was brought forward almost immediately, showing that his statements were only a series of falsehoods. [See page 93, Vol. 1.] We see no reason at present to doubt the experiments of Langstroth and Berlepsch; besides we find Italian workers, in almost all Apiaries of black bees in our vicinity, but the colonies producing them show only black drones.

So long as our object is honey, and we only procure Italians because they gather *more* honey, we cannot see that the question matters particularly either way. Every Apiary of 50 colonies should have at least one Imported Queen, and all colonies should be made with Queens reared from this one. You will then have good bees in every hive without bothering about drones at all. All Apiaries of less than 50 stocks can have an Imported Queen's daughter—as they cost but \$1.00 each, you can buy 4 or 5 if need be, until

you get one whose workers suit you. Every one who rears Queens *for sale* should *certainly* have an Imported Queen. At the low price at which they are now offered there is no excuse for such blundering, and if our readers would decide to patronize none but those having *bona fide* Imported Queen mothers, advertisers would make haste to supply themselves. The plea that Imported Queens do not always produce three banded workers, we think a mistake. All the reports from Queens of the Nunn importation agree without an exception that the workers are not only three banded, but that they are quite superior as honey gatherers. James Bolin's letter just at hand is a sample of one of them.

The idea of judging a Queen by her *looks* and granting a diploma etc., on the strength of it, is simply ridiculous. Concerning Imported Queens, I would say that I obtained one of friend Nunn and if there are any better ones anywhere I would like to know it. Her progeny and those of Queens reared from her are the very best workers I have in my Apiary.

JAMES BOLIN, West Lodi, O. Jan. 7th, '75.

Friend H. had better have sealed drone brood instead of eggs, it can then be sent safely by mail without bees. We can send it at the same price we do eggs from our Imported Queen, viz., 25c for a piece 2×3 inches. We have little faith however that the drones can be made available.

I have a hot-bed and my bees are doing well in it. I have about 60 plants in it, I think they help the atmosphere a great deal.

A. N. DRAPER, Upper Alton, Ills. Feb. 1st, '75.

My boy had when we came here a small box of sun-flower seeds, which he kept as one of his playthings, last spring he accidentally spilt them down in the garden by the fence, and, old as they were, they came up profusely. They looked so thrifty, I took it into my head I would transplant them. I went to work and set them all around in the fence out of the way where there would nothing else grow to advantage, and if you will believe me, I had an enormous crop; and behold when they blossomed the bees went at them in earnest, and after the bees got through with them, there were several quarts of seed. I sold a dollar's worth to my druggist, and the balance I fed out to my hens, and as a writer of old has said, I found nothing so good and nourishing for laying hens as sun-flower seeds. Then I cut off the heads and place them near the bee hives, fill them with sugar and water, and that suits the bees to a T. So you see I was to no expense, and they paid well, I write this that others may be benefitted as well as myself, by so doing.

Common sense, philosophy and religion, alike teach us to receive with becoming reverence, all undoubted facts, whether in the natural or spiritual world; assured that however mysterious they may appear to us they are beautifully consistent in the sight of Him whose "understanding is infinite."

DR. R. HITCHCOCK, South Norwalk, Conn. Feb. 2nd.

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